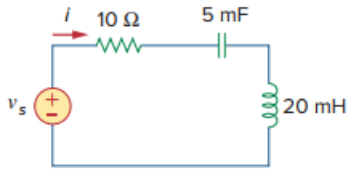


Problem

Find the steady-state current i in the circuit of Fig. 9.42, when $v_s(t) = 115 \cos 200t$ V.

Figure 9.42:



Step-by-step solution

Step 1 of 4

Refer to Figure 9.42 in the textbook.

The sinusoidal expression of source voltage is, $v_s(t) = 115 \cos(200t)$ V

Here,

The magnitude of the voltage v_m is 115 V

The angular frequency ω is 200 rad/s

The phase angle is 0° .

The phasor representation of voltage is,

$$\mathbf{V}_s = 115 \angle 0^\circ \text{ V}$$

Step 2 of 4

Calculate the impedance of 5 mF capacitor.

$$\begin{aligned} \mathbf{Z}_C &= \frac{1}{j\omega C} \\ &= \frac{1}{j(200)(5 \times 10^{-3})} \\ &= -j1 \end{aligned}$$

Calculate the impedance of 20 mH inductor.

$$\begin{aligned} \mathbf{Z}_L &= j\omega L \\ &= j(200)(20 \times 10^{-3}) \\ &= j4 \end{aligned}$$

Step 3 of 4

Determine the equivalent impedance of the series RLC circuit.

$$\begin{aligned} \mathbf{Z}_{in} &= R + \mathbf{Z}_C + \mathbf{Z}_L \\ &= 10 - j1 + j4 \\ &= 10 + j3 \\ &= 10.44 \angle 16.7^\circ \Omega \end{aligned}$$

Calculate the current through the circuit.

$$\begin{aligned} \mathbf{I} &= \frac{\mathbf{V}}{\mathbf{Z}_{in}} \\ &= \frac{115 \angle 0^\circ}{10.44 \angle 16.7^\circ} \\ &= 11.015 \angle -16.7^\circ \text{ A} \end{aligned}$$

Convert the current expression from phasor domain to time domain.

$$\begin{aligned} i &= i_m \cos(\omega t + \phi) \\ &= 11.015 \cos(200t - 16.7^\circ) \text{ A} \end{aligned}$$

Therefore, the steady state current in the circuit is $11.015 \cos(200t - 16.7^\circ) \text{ A}$.